

Response to Reviews: JCGS-25-420

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We thank the anonymous associate editor and reviewers for their careful reviews. What follows is our point-by-point response to reviewer comments.

Note that the original reviewer comments are in normal and *our response is in italic text*.

Reviewer 1

Summary:

This manuscript proposes an algorithm to evaluate nonlinear dimension reduction (NLDR) techniques. The authors identify some characteristics and limitations of NLDR and develop an algorithm to analyze NLDR results' quality of how well they match the original high-dimensional data.

Strengths:

1. This manuscript is well-organized, providing sufficient background knowledge and examples which makes this paper easy to follow.
2. This paper contains sufficient technical details in the introduction of the algorithm.

Weakness:

1. The main concern is whether the proposed algorithm can properly evaluate the NLDR's performance. Using the RMSE plot with different binwidth values (e.g., in Figure 8 and Figure 11) to evaluate NLDR's performance is problematic for not considering the density within each cluster. For NLDR results that have denser clusters, for example layout a,c,g,h in Figure 11, a smaller binwidth needs be used to achieve the same average count

in each bin, which will cause their lines in Figure 11 to be more upper left. However, denser clusters are not worse NLDR results.

Thank you for this thoughtful comment. We agree that the density of clusters can influence the RMSE curves, and that denser clusters are not inherently indicative of poorer NLDR results. The purpose of our diagnostic was not to penalize density per se, but to provide a way to compare the stability of neighborhood structures across different binwidths. In practice, a representation with highly dense clusters will indeed tend to have lower RMSE at small binwidths, shifting its curve to the upper-left of the plot. This is a reflection of how tightly the representation packs points together rather than a judgment of quality on its own.

To address this concern, we have revised the text to clarify that RMSE should not be interpreted in isolation. Instead, it should be considered in conjunction with other diagnostics, such as the proportion of non-empty bins and visual separation of clusters. We emphasize that our framework is intended to highlight trade-offs across multiple metrics rather than to define a single absolute ranking.

We also note in the revision that dense clusters are not inherently “worse” outcomes—rather, they may reflect meaningful biological or structural features of the data. The diagnostic is designed to make these differences visible so that users can interpret them in the context of their application.

2. In Figure 7, as RMSE steadily increases with the binwidth, it’s not clear why the three binwidths 0.03, 0.05, and 0.07 are chosen. In addition, we shouldn’t use a single simulated dataset to determine the default hyperparameters without any real analysis.

Thank you for raising this important point. We agree that in Figure 7, the steady increase of RMSE with binwidth makes it less obvious why specific values of a_1 were selected. Our intent was not to claim that these three binwidths (0.03, 0.05, and 0.07) are universally optimal, but rather to use them as illustrative examples for exploring trade-offs between RMSE, bin density, and the proportion of non-empty bins. We chose these values because they represent a range of small-to-moderate binwidths that highlight changes across the diagnostic plots.

We also agree with your concern that hyperparameters should not be set based on a single simulated dataset. To clarify, our goal here was to demonstrate the process of tuning rather than to prescribe default hyperparameters.

3. For several figures that contains results from multiple NLDR algorithms, it would be nice to add the information of which algorithm/ what hyperparameter values for each of the subfigure. Many readers may have tried several NLDR algorithms themselves, and they will benefit from those figures to gain insights about behaviors of different NLDRs. Also it’s nice to mention which data is used to generate each figure (e.g., in Figure 7).

Thank you for this helpful suggestion. We agree that including the NLDR algorithm names, hyperparameter settings, and dataset details in the figure panels will improve clarity and make

the results more reproducible and interpretable for readers. We have revised the manuscript accordingly:

- *Each subfigure now includes labels indicating the algorithm used and its key hyperparameter values.*
- *Figure captions have been expanded to explicitly state the dataset used for generating the results.*
- *For figures with multiple algorithms, the layout now clearly shows which configuration corresponds to each subfigure.*

4. In section 3.6.1, it says “Values of b_1 between 2 and $b_1 = \sqrt{n/r_2}$ are allowed”, why the choice of b_1 should consider r_2 ?

These are the reasons:

- b_1 *controls the number of bins along the x-axis (the first NLDR dimension).*
- *In a hexagonal grid, the spacing along the y-axis (second dimension) is not independent. It’s linked to the x-direction because hexagons “stack” with vertical offsets.*
- *The vertical extent is determined by the hexagon height, which in turn depends on the chosen bin width along the x-direction.*

So, to ensure the grid actually fits within the data range in both directions, the allowable values of b_1 are constrained by r_2 , the range of the second dimension. If b_1 is too large, the hexagons become too narrow/tall, and the grid will not cover the vertical range properly.

That’s why the condition is written as:

$$2 \leq b_1 \leq \sqrt{\frac{n}{r_2}}$$

tying the number of x-bins (b_1) to the vertical range of the data (r_2).

Reviewer 2

Summary

The manuscript proposes a novel, mostly graphical diagnostic to evaluate nonlinear dimension reduction (NLDR) layouts by effectively “lifting” the 2-D embedding back into the original high-dimensional space and inspecting the fit visually via tours (i.e., sequences of 2D-linear projections of the original high-D space). The goal is to help analysts discern which 2-D projection is the most “honest” (faithful) representation of the true structure in the data. The paper’s contributions are:

1. an algorithm that overlays a geometric model on an NLDR layout using hexagonal binning and Delaunay triangulation in the embedding, then maps this model into the original large-D data space for visual assessment;
2. a new goodness-of-fit metric, the RMSE in original space, computed as the root mean squared distance between each point and its bin centroid in high-D (Eq 2) (3) interactive visualization tools (implemented in published software) to compare/investigate different NLDRs

It is mostly well written, addresses an important problem with critical real-world relevance, and provides a very well done, extensively documented R implementation. However, this reviewer is skeptical about the suitability of the newly proposed GOF metric to choose the most “reasonable” or “faithful” NLDR, finds the empirical evaluation of it to be severely lacking without comparison to alternative measures from the literature, and would request some restructuring and shortening of the paper.

Strengths

Relevance, Accessibility, and Intuitiveness

The way this method visualizes the spatial distortions induced by NLDR seems uniquely accessible to me, and, coupled with the modern interactive/animated visualizations implemented for it, seems likely to become a very useful tool to investigate and understand high-D data structures while avoiding over-reliance on / misinterpretation of misleading 2D patterns so prevalent in many areas at the moment (e.g. Izarry (2024), Chari & Pachter (2023)).

Background Material & Examples

The paper is fully fleshed out with illuminating examples, and I commend the authors for the effort they put into documenting their package and providing very informative videos etc of the method in action.

Reproducibility and Code Availability

The paper is exemplary in this regard - the underlying software is a CRAN compliant R package, and the Github repository itself allows to recompile the entire paper and appendix with all figures etc from scratch.

Major Issues

No Standard Embedding Quality Metrics or Comparisons

My primary concern is the absence of established quantitative metrics

1. to evaluate to what extent the newly proposed “RMSE” metric corresponds to established notion of embedding faithfulness and
2. to validate the different embeddings under consideration.

The paper’s only evaluation of “honesty” is done via the proposed custom “RMSE” in high dimensional space (defined in Section 3.4) and visual heuristics, but it does not report any conventional measures like trustworthiness/continuity scores, k -Neighborhood preservation rates (as derived from the co-ranking matrix - e.g. R_{NX} -curves for different neighborhood sizes k , c.f. (Lee et al, 2015), (Lueks et al, 2011)), or Shepard diagrams of original space vs. embedding space distances. This is a significant omission because NLDR almost always faces a trade-off between optimizing preservation of local structures vs preservation of global structures, which these metrics are (mostly) designed to resolve / analyze. For example, the authors could have computed such measures of trustworthiness for each NLDR layout to see if the one with lowest “RMSE” indeed had the highest (local and or global) neighborhood structure preservation but no such analysis was provided. The manuscript doesn’t even mention or discuss these metrics, which is not acceptable in a study focused on faithful representation, in my opinion.

Questions a revised version should aim to resolve:

1. does “RMSE” measure local or global structure preservation (or a mixture of both)? From the definition, it seems to be focused on how well small neighborhoods in embedding space (the bins) correspond to small neighborhoods in high-D space, with, IIUC, particularly high values if there are lots of “hard intrusions” (i.e., points that are embedded in the same 2D bin despite being far apart in high-D space)?
2. how strongly are “RMSE” values correlated to other established measures of (local or global) structure preservation? and (how) are these correlations affected by the binning hyperparameters?
3. are the NLDRs selected by lowest “RMSE” the ones that would also be selected by other, more established measures of embedding faithfulness whose characteristics are well understood? at the very least, such measures should be provided, compared and discussed for all the embeddings under consideration in the case studies.

The reliance of “RMSE” on euclidean distances in high-D space also seems limiting:

1. at least for coarser bins and/or highly non-linear data structures, geodesic distances along the data manifold à la ISOMAP are likely to measure a more relevant kind of dissimilarity

2. a definition based on a more general notion of distance would make this applicable to investigating 2D embeddings of data types for which the euclidean distance of their numerical data representations is not a suitable/applicable measure of dissimilarity (e.g. images, text).

Insufficient Validation of the Proposed Diagnostic (RMSE) Across Methods

Related to the above points - the paper has no theoretical or empirical demonstration that minimizing this “RMSE” aligns with minimizing other distortion measures like stress or maximizing neighborhood preservation (as measured by the co-ranking matrix). Furthermore, the claim that RMSE can be used to compare different NLDR methods on equal footing deserves scrutiny. Different methods optimize different criteria; it’s plausible that one method might (tends to) achieve lower “RMSE” simply by construction, especially if my reading that it mostly rewards embeddings that manage to avoid hard intrusions is correct. Overall, much more critical validation of the RMSE diagnostic is needed to convince readers that “honesty” as defined here aligns with meaningful goodness-of-fit in embedding.

Limitations of the Evaluation (Simulation Study Design):

The manuscript includes a simulation to illustrate the approach (the “2NC7” dataset with two nonlinear clusters in 7-D) and two real-data applications (single-cell RNA-seq and an image dataset of handwritten digits). While these examples are appropriate, the scope of simulation scenarios is narrow, and the analysis lacks any notion of variability. The simulation seems to be a single realization of a specific two-cluster scenario. The authors did not explore, for instance, varying the overlap between clusters, adding more clusters or different manifold shapes, or introducing different noise levels. Ideally, the authors would systematically vary factors like: number of clusters or pattern types, intrinsic dimensionality of structures, sample size, etc., and show that their approach reliably selects the truth (or at least, the most interpretable view) in each case, and how well alternative measures would do so. Because only one run is shown, there is also no information how stable the RMSE curves in Figures 8 and 13 are across replications. I encourage the authors to perform more systematic, replicated simulations, report measures of variability for their performance evaluations, compare against relevant alternatives, and take much more care not to overstate their conclusions beyond the tested scenarios (e.g. the claim that it “identify bad representations so they can be avoided” is based on a very limited set of examples).

Minor Issues

1. First part of the title seems inappropriate for a scientific paper to me, as does the implied (and not substantiated, IMO) claim that the proposal is (primarily) useful to reliably

select one of many NLDRs (as opposed to primarily useful for visually investigating details about a specific 2D NLDR).

2. For marketing and clarity reasons, I would strongly suggest to not name the metric “RMSE”, but something more distinctive/descriptive.
3. Flow from 3.1 to 3.2 (with too many sub-subsections) is difficult. Sections 3.2.1 (Scale the data) and 3.2.2 (Construct hexagon grid) are just short paragraphs, would be clearer if combined into a single numbered list of steps for the model construction. Much of 3 might be clearer and more compact if simply replaced by a direct step-by-step description of the algorithm in algorithmic pseudocode form with suitable short explanations.
4. Section 3.5 is a claim without supporting evidence – many out-of-sample embedding methods exist (some based on rather similar ideas, which should be referenced), you would need to show that this particular one actually works reasonably well for this to deserve space in the paper. I would strongly suggest to simply mention this in the discussion section as a possible avenue for further development instead.
5. Section 3.6 should be drastically shortened and details moved into an appendix.
6. p. 18: “A particular pattern that we commonly see is that analysts tend to pick layouts with clusters that have big separations between them.” The literature/internet contains many lively discussions of the perils of mis-interpreting and post-hoc’ing NLDRs (e.g. (Izarry, 2024), (Chari & Pachter, 2023)), it might be good to cite/summarize these instead of/in addition to basing your motivation on personal anecdotal evidence? The tone here seems overly conversational to me (also: “we almost always see there are no big separations in the data”) and should be made more formal.
7. Section 4, esp. p.19: IMO, “to compare and assess a range of representations” an analyst really should look at their respective RNX curves, stress values, and Shepard diagrams, think about whether they care more about local or global structure preservation, and then decide based on those factors. The method proposed here seems to me to be mostly useful to then investigate specific regions / slices of high-D space to see for which sets of data points the high-D structure is (not) preserved well in the NLDR. I find this whole section to be vastly overselling the proposal – it ignores and implicitly discards most of the prior work in this field. Its intro paragraphs are also highly redundant with other content in the paper and should be cut.
8. Section 5 should be moved into an appendix to shorten the paper somewhat. Figure in Section 5.1. could benefit from coloring the clusters in different colors so global structure preservation is also somewhat legible from the figures for the NLDRs (e.g. “is orange cluster between green and blue as in high-D or do they switch positions”). Section 5.2 might also have benefited from a corresponding ISOMAP or similar embedding of this data suitable for truthfully (isometrically!) unrolling the high-D manifold, and then showcasing that such a successful “unrolling” can be discerned from the way the

projections look? tSNE just isn't isometric, so the distortions visible here are entirely expected, IMO?

Typos and grammar:

Revised version should be much more carefully proofread: “appropraite”, “doen’t”, “it is has two separated nonlinear clusters”, “the methods tNSE”

We appreciate your careful attention to detail and for pointing out the typographical and grammatical errors. We have carefully proofread the entire manuscript and corrected the issues you identified (e.g., “appropraite” → “appropriate,” “doen’t” → “doesn’t,” “hastwo” → “has two,” “tNSE” → “tSNE”), along with several other minor typographical inconsistencies. The revised version has been thoroughly checked to ensure clarity and correctness.

Loss function values

tSNE optimizes a Kullback–Leibler divergence between high- and low-dimensional pairwise similarities; however, computing the full loss exactly is feasible but expensive, and standard implementations often use approximations for efficiency. UMAP minimizes a cross-entropy loss over a fuzzy simplicial set, but the true loss cannot be fully tracked because stochastic gradient descent with limited negative sampling approximates the repulsive forces, making the exact loss both computationally prohibitive and noisy. PHATE does not explicitly calculate a conventional loss function; its embedding is derived from diffusion potential distances and MDS, so there is no meaningful per-epoch loss to report. TriMAP optimizes a loss based on triplet constraints, but the loss is generally only evaluated over sampled triplets rather than all possible triplets, meaning the full loss is not computed. PaCMAP, in contrast, reports an actual loss function during optimization, using a fixed set of negative samples for repulsion, making it possible to monitor per-epoch loss values reliably, though this estimate may be biased relative to the theoretical full pairwise objective.

Metrics

1. k -Neighborhood Preservation Rates / R_{NX} -curves

Purpose: Measures how well local neighborhood structures are preserved by a dimensionality reduction method.

Setup:

- You have high-dimensional data $X = \{x_1, \dots, x_n\}$ and a lower-dimensional embedding $Y = \{y_1, \dots, y_n\}$.

- For each point x_i , define its set of k nearest neighbors in high-dimensional space as $N_k(x_i)$.
- Similarly, define $N_k(y_i)$ in the embedding space.

Neighborhood preservation for a single point:

$$R_{NX}^{(i)}(k) = \frac{|N_k(x_i) \cap N_k(y_i)|}{k}$$

- Counts the fraction of neighbors in the high-D space that are still neighbors in the embedding.

Overall k -Neighborhood preservation rate:

$$R_{NX}(k) = \frac{1}{n} \sum_{i=1}^n R_{NX}^{(i)}(k)$$

- This gives a single number per neighborhood size k .
- Plotting $R_{NX}(k)$ against k gives the R_{NX} -**curve**, showing local-to-global structure preservation.

Reference: Lee et al., 2015; Lueks et al., 2011.

Interpretation:

- $R_{NX}(k) = 1 \rightarrow$ perfect preservation of k -neighborhoods.
- $R_{NX}(k) \approx 0 \rightarrow$ neighborhoods are poorly preserved.

2. Shepard Diagrams

Purpose: Evaluates how well distances between points in high-dimensional space are preserved in the embedding.

Setup:

- For all pairs of points x_i, x_j , compute high-dimensional distances $d_X(i, j)$.
- In the embedding, compute distances $d_Y(i, j)$.

Shepard diagram:

- Plot each pair $(d_X(i, j), d_Y(i, j))$ as a point in 2D.
- Often a smooth curve is added to visualize the trend.

Interpretation:

- Points lying close to the diagonal $d_Y = d_X \rightarrow$ good preservation of distances.

- Large deviations $\rightarrow$ distortion.
- Can quantify with **stress** or **Pearson/Spearman correlation** between d_X and d_Y .

Applications:

- Visualizes global distortions.
- Complements k -neighborhood preservation (local structure).

3. KNN Accuracy

Intuition:

If an embedding preserves local neighborhoods well, then points close together in high-dimensional space should remain close in the low-dimensional embedding. A simple way to check this is to see how well a **K-Nearest Neighbors classifier** can predict labels in the embedding space.

Mathematical Definition:

1. Let $Y = \{y_1, \dots, y_n\}$ be the low-dimensional embedding of high-dimensional data $X = \{x_1, \dots, x_n\}$.
2. Let $L = \{l_1, \dots, l_n\}$ be the true labels for each observation.
3. For each point y_i , find its k nearest neighbors $\text{NN}_k(y_i)$ in the embedding space.
4. Predict y_i 's label as the **majority vote** among neighbors:

$$\hat{l}_i = \text{mode}(\{l_j : j \in \text{NN}_k(y_i)\})$$

5. KNN accuracy is:

$$\text{KNN Accuracy} = \frac{1}{n} \sum_{i=1}^n \mathbf{1}\{\hat{l}_i = l_i\}$$

- $\mathbf{1}\{\cdot\}$ is the indicator function.

Notes:

- This is a **leave-one-out cross-validation (LOO-CV)**.
- For larger datasets, you can use **k-fold CV** to reduce runtime.

4. SVM Accuracy (Local Evaluation)

Intuition:

SVM Accuracy measures local structure preservation in a **more flexible way**. It uses a kernel (e.g., RBF) to capture nonlinear separations in the embedding space. If the embedding preserves neighborhoods, an SVM trained on some points should correctly classify held-out points.

Mathematical Definition:

1. Let Y be the embedding and L the labels.
2. Partition Y into K folds: $Y = Y_1 \cup \dots \cup Y_K$.
3. For each fold k :
 - Train an SVM on $Y \setminus Y_k$ (all other folds).
 - Predict labels on Y_k .
4. SVM Accuracy is the overall proportion of correct predictions:

$$\text{SVM Accuracy} = \frac{1}{n} \sum_{i=1}^n \mathbf{1}\{\hat{l}_i = l_i\}$$

- Often an **RBF kernel** is used:

$$K(y_i, y_j) = \exp(-\gamma \|y_i - y_j\|^2)$$

- To speed up large datasets, one can approximate the kernel with **Nyström method** (low-rank approximation).

Suggested metric by us

For each bin h with n_h points:

$$SS_h = \sum_{i=1}^{n_h} \sum_{j=1}^p (x_{hij} - C_{hj}^{(p)})^2$$

- $x_{hij} = j^{th}$ variable of the i^{th} point in bin h
- $C_{hj}^{(p)} = j^{th}$ coordinate of the bin center C_h
- p = number of dimensions

The **total within-bin sum of squares** is:

$$WBSS = \sum_{h=1}^m SS_h$$

- Measures **how spread out points are around their assigned bin center**.
- Smaller WBSS $\rightarrow$ points are tightly clustered within bins $\rightarrow$ embedding preserves local structure better **within the bin**.

Strengths

- Directly reflects the **fidelity of the embedding to your “bin model”**.
- Easy to compute in high-dimensional space.
- Conceptually similar to your RMSE.

Limitations

- **Does not capture neighborhood preservation across bins**, i.e., whether neighbors in high-dimensional space remain neighbors in low-dimensional space.
- **Does not reflect global structure preservation**.